Experiment - 04

Data Modelling and Form Design for a Vehicle Service Request Form in Zoho Creator

Name: Shrivarthini I

REG NO: 241401100
Programme: B.Tech CSBS

AIM

To design and build a functional **“Vehicle Service Request”** application in Zoho Creator that demonstrates the use of Basic Fields available in the Form Builder and models the data structure of a vehicle servicing workflow through a properly configured service request form, an auto-generated report view, and an Entity-Relationship (ER) diagram.

PROCEDURE

1. Application and Form Creation

- Logged into Zoho Creator and created a new application named **“Vehicle Service & Repair Request Management”**.
- Inside the application, opened the Form Builder and created the **Service Request Form**.
- The form was designed to capture vehicle owner details, contact information, vehicle information, service requirements, and request details.
- Zoho Creator automatically generated a linked report for viewing and managing submitted service requests.

2. Drag-and-Drop of Basic Fields

The Basic Fields palette was expanded on the left panel. Required fields were dragged and dropped onto the form canvas and configured through the Field Properties panel.

Basic Field	Field Name	Configuration

Name	Owner Name	First Name and Last Name enabled; field marked Mandatory and treated as personal data.
Single Line	Vehicle Registration Number	Used to capture the vehicle registration number; marked Mandatory.
Drop Down	Vehicle Type	Options such as Car, Bike, SUV, Van and Other.
Email	Email Address	Used for customer communication with built-in email validation.
Phone	Contact Number	Used to capture the owner's contact number with phone validation.
Address	Pickup Address	Used to capture the vehicle pickup location.
Number	Odometer Reading	Used to record the current vehicle odometer reading in kilometres.
Drop Down	Service Type	Options such as General Service, Oil Change, Brake Service, Engine Repair, AC Service and Other.
Multi Line	Issue Description	Used to describe the vehicle problem or service requirement.
Date	Preferred Service Date	Used to select the customer's preferred service date.
Time	Preferred Service Time	Used to select the preferred service time.
Radio	Service Priority	Options such as Normal, Urgent and Emergency.

3. Form Sections

The Service Request Form was organized into the following sections:

Owner Details

- Owner Name
- Vehicle Registration Number
- Vehicle Type

Contact Details

- Email Address

- Contact Number
- Pickup Address

Vehicle & Issue Details

- Odometer Reading
- Service Type
- Issue Description

Service Schedule

- Preferred Service Date
- Preferred Service Time
- Service Priority

4. Configuring Field Properties

- Selected the **Owner Name** field and enabled the required First Name and Last Name display options.
- Marked the Owner Name field as **Mandatory**.
- Configured the Vehicle Registration Number field as a mandatory single-line field.
- Configured Vehicle Type as a Drop Down field containing appropriate vehicle categories.
- Configured Email Address with built-in email validation.
- Configured Contact Number as a Phone field.
- Configured Pickup Address using the Address field and its available address components.
- Configured Odometer Reading as a Number field.
- Configured Service Type as a Drop Down field.
- Configured Issue Description as a Multi Line field.
- Configured Preferred Service Date and Preferred Service Time using Date and Time fields.
- Configured Service Priority using Radio buttons.

5. Submitting a Sample Record and Reviewing the Report

A sample vehicle service request was entered through the live form.

Sample Record:

- **Owner Name:** Ramesh
- **Vehicle Registration Number:** TN 01 AB 1234
- **Vehicle Type:** Car
- **Email Address:** ramesh@example.com
- **Contact Number:** +91 98765 43210
- **Pickup Address:** Chennai, Tamil Nadu

- **Odometer Reading:** 45200 km
- **Service Type:** General Service
- **Issue Description:** Periodic maintenance and engine oil replacement required.
- **Preferred Service Date:** 20th August 2026
- **Preferred Service Time:** 10:00 AM
- **Service Priority:** Normal

After submission, the generated service request report was opened in list view. The submitted record was verified and the individual record was opened to confirm that all entered values were correctly stored.

6. Data Modelling

An Entity-Relationship diagram was designed to document the underlying data model of the Vehicle Service Request application.

The central **SERVICE_REQUEST** entity is associated with the **OWNER**, **VEHICLE**, **CONTACT**, **ADDRESS**, **SERVICE_TYPE**, **SERVICE_SCHEDULE**, and **SERVICE_STATUS** entities.

A vehicle can have multiple service requests over time, while each service request belongs to one vehicle and one owner. Service type and service status are represented as reference entities to support consistent service management.

OUTPUT

The following screenshots represent the completed application:

1. Form Builder View — Owner Name Field Configuration

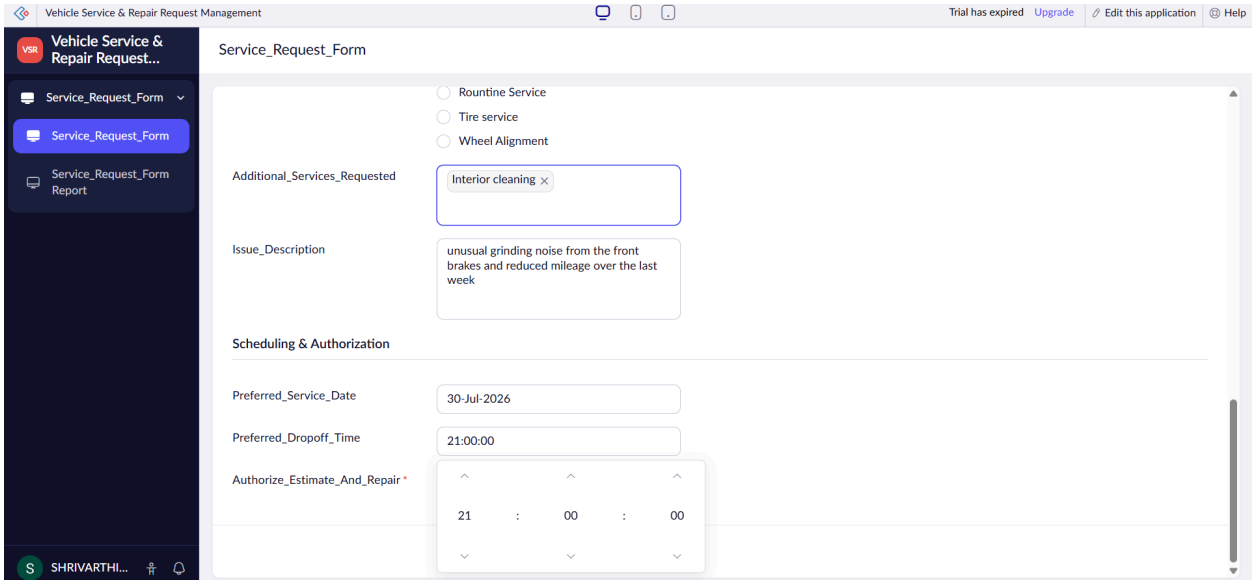


Fig 1. Service Request Form — Basic Fields palette with the Owner Name field properties open.

2. Form Builder View — Vehicle and Contact Details

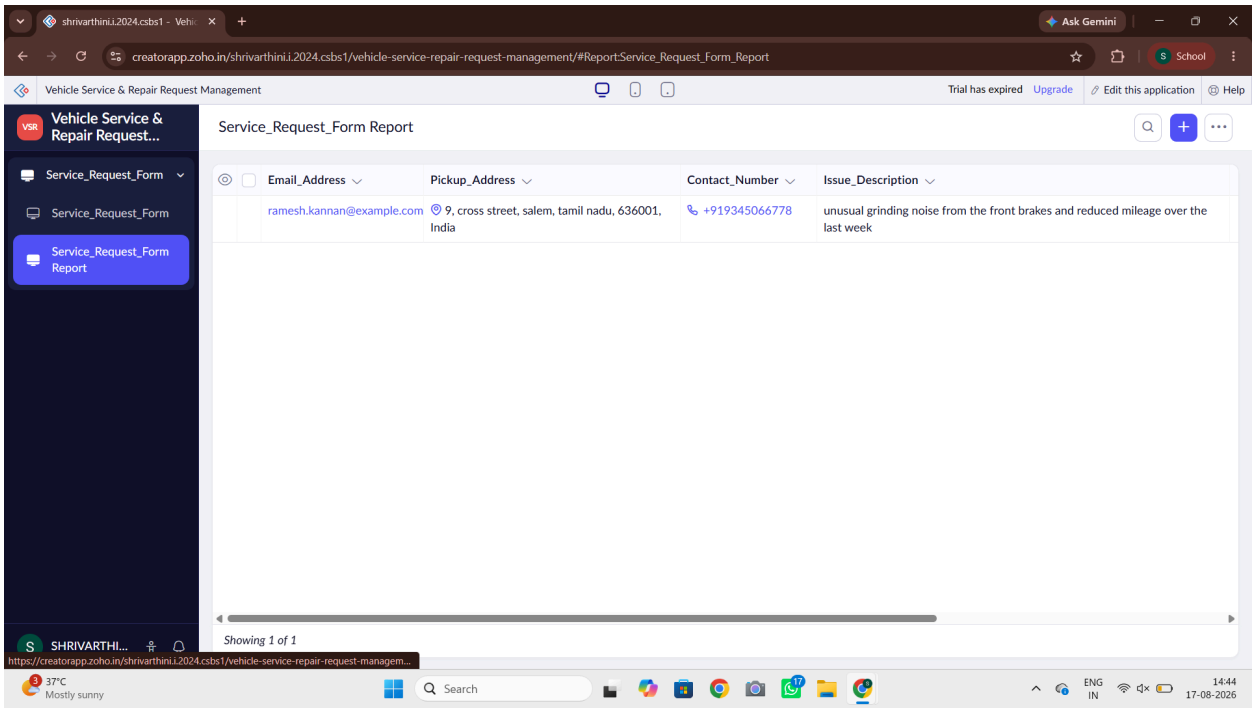
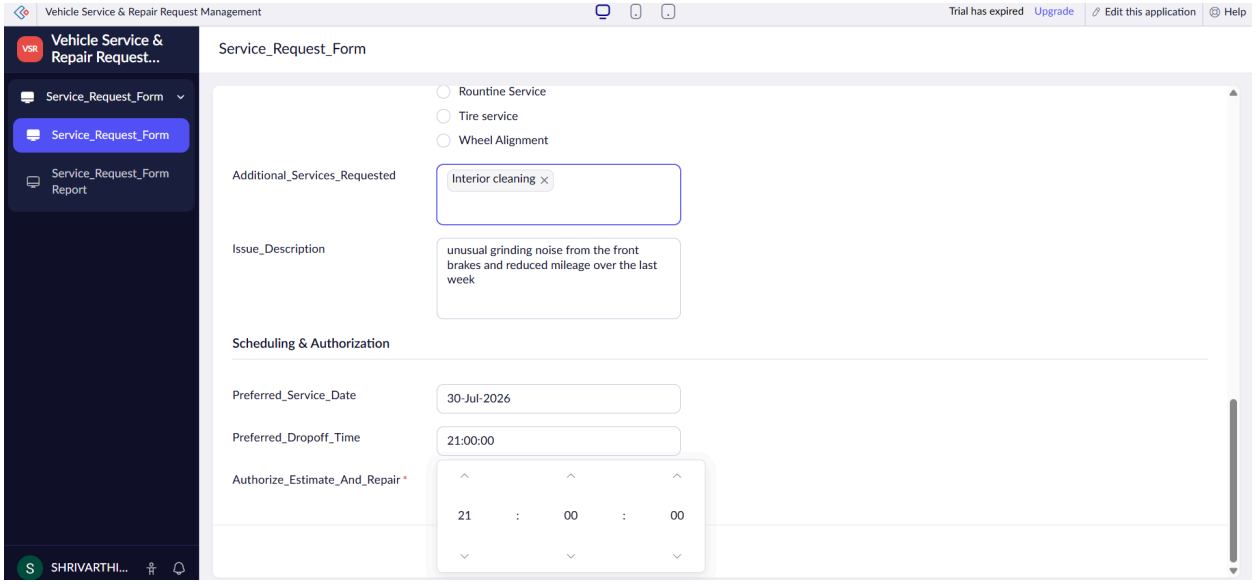


Fig 2. Service Request Form — Vehicle Registration Number, Vehicle Type, Email Address, Contact Number and Pickup Address fields.

3. Report View — Tabular List of Service Requests



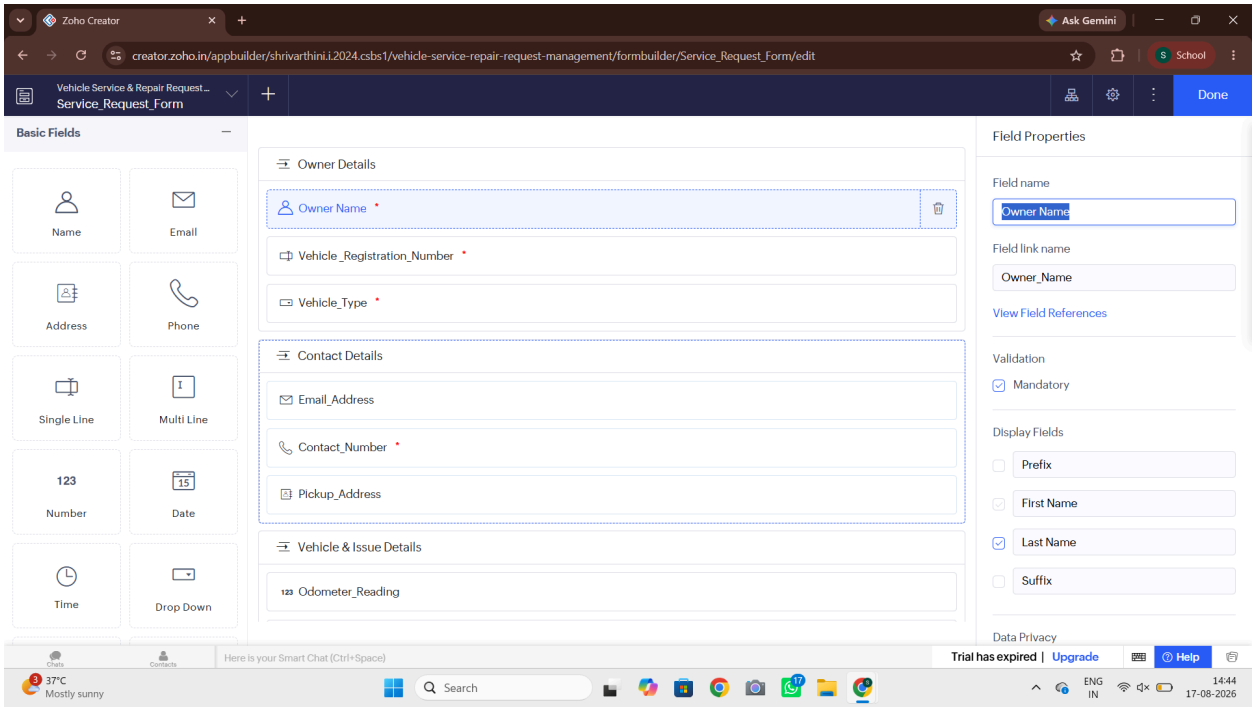


Fig 3. Vehicle Service Requests — grid view displaying the submitted service request.

4. Report View — Individual Service Request

Fig 4. Vehicle Service Request — detail view showing the complete submitted record.

5. Data Model — Entity-Relationship Diagram

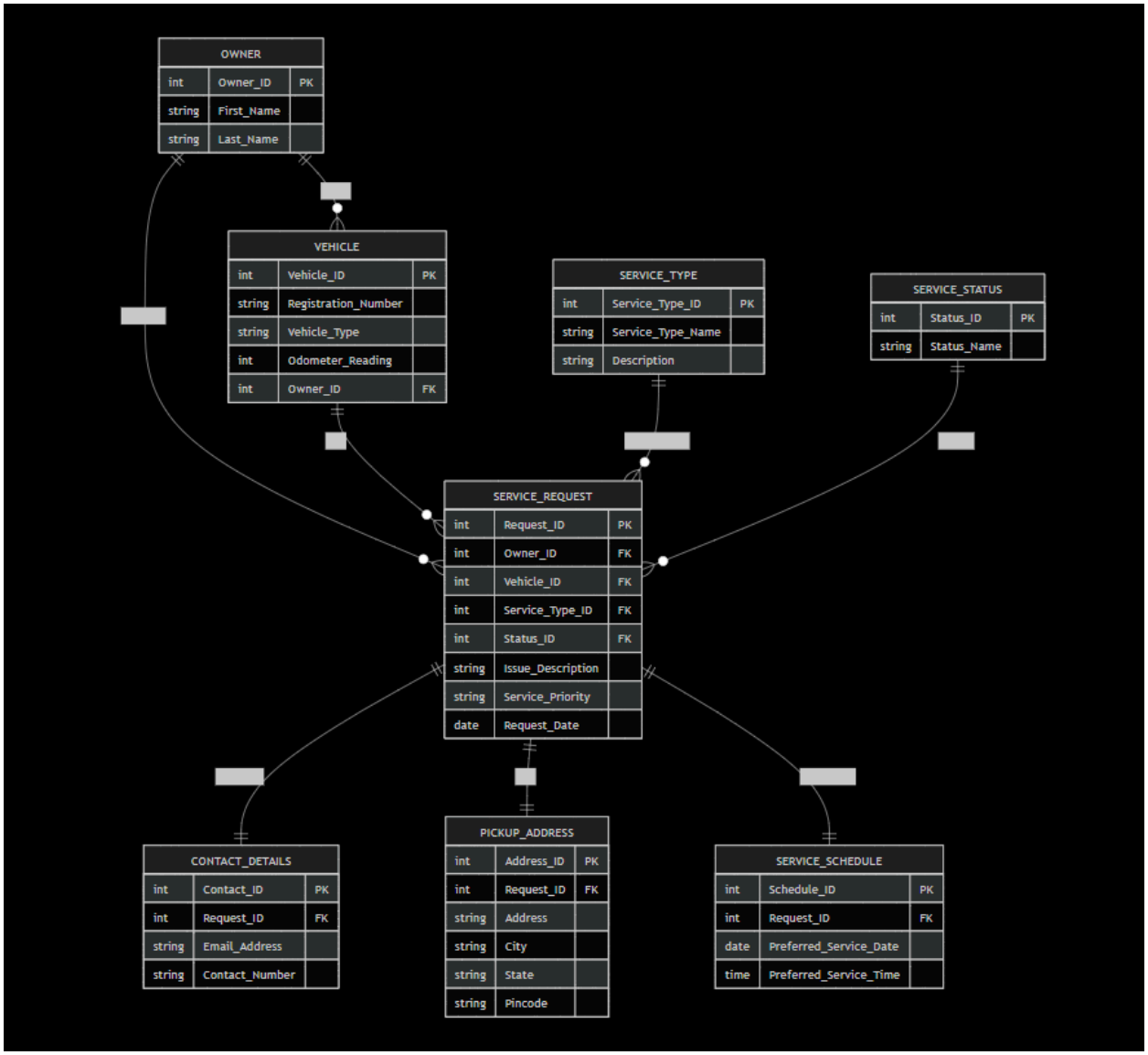


Fig 5. Entity-Relationship diagram representing the Vehicle Service Request data model.

RESULT

A functional **Vehicle Service Request** application was successfully designed and modelled in Zoho Creator. The Service Request Form captures vehicle owner information, contact details, vehicle information, service requirements, and scheduling details using appropriate Basic Fields. The generated report correctly displays the submitted service request, while the accompanying ER diagram represents the relationships between owners, vehicles, service requests, service types, schedules, and service status.